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Trussed ramp

Abstract

A ramp includes an extruded beam extending between a first end and a second end. The beam includes a first plate and a second plate opposite the first plate. A first side wall and a second side wall each extend between the first plate and the second plate. A first brace wall extends from a first corner to the first plate. The first corner is between the second plate and the first side wall. A second brace wall extends from a second corner to the first plate. The second corner is between the second plate and the second side wall.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority of U.S. Provisional Patent Application No. 63/052,677, filed on Jul. 16, 2020, and claims priority of U.S. Provisional Patent Application No. 63/143,497, filed on Jan. 29, 2021. The contents of which are both incorporated by reference herein in their entireties.

FIELD

(1) The present disclosure relates to cargo loading and transportation and specifically to ramps configured for loading and unloading trucks and trailers.

BACKGROUND

(2) Ramps allow vehicles and cargo to be loaded onto open or enclosed trailers. Load levelers support loads above open drop deck semi-trailers. U.S. Pat. Nos. 7,930,788 and 8,347,439 disclose examples of combination ramp and load levelers. U.S. Pat. No. 8,448,278 discloses examples of a ramp. U.S. Pat. No. 8,061,943 discloses examples of a load leveler. Each of U.S. Pat. Nos. 7,930,788; 8,061,943; 8,347,439; and 8,448,278 are incorporated herein by reference in their entireties.

(3) This Background is intended to introduce various aspects of the art, which may be associated with the present disclosure to thereby assist in providing a framework to facilitate a better understanding of particular aspects of the present disclosure. Accordingly, it should be understood that this Background should be read in this light, and not necessarily as admissions of prior art.

BRIEF DISCLOSURE

(4) An example of a ramp includes an extruded beam extending between a first end and a second end. The beam includes a first plate and a second plate opposite the first plate. A first side wall and a second side wall each extend between the first plate and the second plate. A first brace wall extends from a first corner to the first plate. The first corner is between the second plate and the first side wall. A second brace wall extends from a second corner to the first plate. The second corner is between the second plate and the second side wall.

(5) In examples, the first plate is a top plate and the second plate is a bottom plate. The first corner is a corner between the first sidewall and the bottom plate and the second corner is a corner between the second sidewall and the bottom plate. The first and second brace walls meet at the midpoint of the top plate. The first side wall includes a tongue that extends lengthwise along the first side wall outwardly away from the second side wall. The second side wall includes a groove extending lengthwise along the second side wall. The tongue has a first dimension to fit within a second dimension of the groove.

(6) In additional examples, the center wall extends between the first plate and the second plate, wherein the center wall is equidistant between the first side wall and the second side wall. The center wall is parallel to the first side wall and the second side wall. A third brace wall connects to the first plate and a fourth brace wall connects to the first plate. The third brace wall extends from a third corner to the first plate, the third corner being between the center wall and the second plate. The fourth brace wall extends from a fourth corner to the first plate, the fourth corner being between the center wall and the second plate. The first brace wall connects to the third brace wall at the first plate and the second brace wall connects to the fourth brace wall at the first plate.

(7) In further examples, the first end is orthogonal to the first plate and the second end is orthogonal to the first plate. A first end piece is secured about the first end and engages at least one of the first plate or the second plate. A second end piece is secured about the second end and engages at least one of the first plate or the second plate. The first end piece may be a first end hook and the second end piece may be a second end hook. A textured surface may be secured to the first plate. The first plate may include grooves that define ribs and the grooves further define dovetails into the ribs. The textured surface may include an adhesive or an epoxy, and the adhesive or epoxy solidifies within the dovetails to form a mechanical connection between the textured surface and the first plate.

(8) In an example of a ramp, the ramp includes first and second extruded beams that extend between a first end and a second end. Each beam includes a first plate having an exterior textured surface and a second plate opposite the first plate. Each beam includes a groove wall extending between the first plate and the second plate. The groove wall includes a groove extending into the groove wall lengthwise along an outside of the groove wall. Each beam includes a tongue wall that extends between the first plate and the second plate. The tongue wall includes a tongue extending lengthwise along the first side wall outwardly from the tongue wall in a direction away from the groove wall. Each beam includes a first brace wall that extends from a first corner to the first plate, the first corner being between the second plate and the first side wall. Each beam includes a second brace wall that extends from a second corner to the first plate, the second corner being between the second plate and the second side wall. A first end piece engages across the first ends of the first and second extruded beams. A second end piece engages across the second ends of the first and second extruded beams. The tongue of the first extruded beam is received within the groove of the second extruded beam to connect the first extruded beam to the second extruded beam in a lengthwise manner. The first extruded beam includes a first chamfered exterior corner where the tongue wall

of the first extruded beam connects to the second plate of the first extruded beam. The second extruded beam includes a second chamfered exterior corner where the groove wall of the second extruded beam connects to the second plate of the second extruded beam. A weld joint secures the first extruded beam to the second extruded beam. The weld joint is located between the first chamfered exterior corner and the second chamfered exterior corner.

(9) In additional examples of the ramp, the first and second extruded beams each further include a center wall, a third brace wall, and a fourth brace wall. The center walls extend between the first plate and the second plate and are equidistant between the respective groove wall and the tongue wall. The third brace wall extends from a third corner to the first plate, the third corner being between the center wall and the second plate. The fourth brace wall extends from a fourth corner to the first plate, the fourth corner being between the center wall and the second plate. The first brace wall connects to the third brace wall at the first plate and the second brace wall connects to the fourth brace wall at the first plate. The first end piece includes first shelf that engages the second plate of the first and second extruded beams and the second end piece includes a second shelf that engages the second plate of the first and second extruded beams. The exterior textured surfaces of the first and second extruded beams is a continuous textured surface across the respective first plates of the first and second extruded beams.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is described with reference to the following Figures. The same numbers are used throughout the Figures to reference like features and like components.
- (2) FIG. 1 is a perspective view of an example of an apparatus within the present disclosure.
- (3) FIG. 2 is a perspective view of a beam of the apparatus of FIG. 1.
- (4) FIG. 3 is a cross-sectional view of a beam in an example of the apparatus.
- (5) FIG. 4A is a detailed view of a top plate as denoted by line 4-4 in FIG. 3.
- (6) FIG. 4B depicts the detailed view of FIG. 4A with a textured surface.
- (7) FIG. 5 is a detailed magnified view of a joint between two beams in an example of the apparatus.
- (8) FIG. 6 is a side view of an example of an end piece for use with an example of the apparatus.
- (9) FIG. 7 is a side view of an example of a further end piece for use with an example of the apparatus.
- (10) FIG. 8 is a perspective view of an example of the apparatus within the present disclosure.
- (11) FIG. 9 is a perspective view of a further example of an apparatus within the present disclosure.
- (12) FIG. 10 is a cross-sectional view of the apparatus taken along line 10-10 of FIG. 9.
- (13) FIG. 11 is a cross-sectional view of the apparatus taken along line 11-11 of FIG. 9.

DETAILED DISCLOSURE

(14) FIGS. 1-8 depict examples of the apparatus 100 as disclosed herein. The apparatus 100 is exemplarily a ramp. The apparatus 100 is constructed of one or more beams 102 arranged side-by-side and extend for the length dimension L of the apparatus 100. The side-by-side arrangement of the beams 102 will be described in further detail herein. The apparatus 100 includes a ramp end 103 and a hook end 104 secured, for example, by welding or adhesive, across the ends of the beams 102. While the apparatus 100 is depicted with two different ends 103, 104, in other examples, the ends 103, 104 can be symmetrical, thus making the overall apparatus symmetrical across both the length dimension and the width dimension.

(15) Each beam 102 may include a textured surface 106. The textured surface 106 may be a grit surface applied with an adhesive. The grit surface may be an acrylic, epoxy, or polymer coating that contains grit and is applied to the surfaces of the beams 102. The textured surface 106 may also be

formed in the beam **102** as part of the extrusion process or may be machined or chemically etched into the surface of the beam **102**. While not depicted in the apparatus of FIG. 1, the textured surface may further include rods or bars welded across one or more beams in the width dimension.

(16) FIG. 2 provides a view of the apparatus **100** without the ends **103**, **104**. In this view the interior structure of the beams **102** can be seen as well as that the beams **102** are interconnected along the length dimension (L) at the adjacent sides of the beams **102**. In an example, the apparatus **100** is exemplarily constructed of four adjacently aligned beams **102**. In a still further example, each of the beams **102** is 9 inches in the width dimension for an overall width of the apparatus **100** of 36 inches.

(17) FIG. 3 is a detailed end view of an example of a beam **102**. The beam **102** includes a top plate **108** which may include a surface texture as described above. In a still further example, the top plate **108** of the beam includes texturing, for example, ribs **110** that are adapted to form a bond with the aforementioned textured coating comprised of e.g. acrylic, epoxy, or polymer materials. The beam **102** further includes a bottom plate **112**. It will be recognized that the beam **102** can be manufactured as an extrusion, for example of aluminum. In such an example, the view shown in FIG. 3 may be the shape of such an extrusion.

(18) The beam **102** differs from other beams as described herein in a number of ways. The beam **102** includes opposed side walls, however, a tongue **114** extends from a tongue wall **116** and a groove **118** extends into a groove wall **120**. The tongue wall **116** and groove wall **120** extend the length of the beam **102** in the length dimension. The tongue **114** extends for the length of the tongue wall **116** and the groove **118** extends for the length of the groove wall **120**. The tongue wall **116** connects to the bottom plate **112** at a chamfered corner **122** and the groove wall **120** connects to the bottom plate at a chamfered corner **122**. These features will be described in further detail subsequently herein.

(19) The beam **102** further includes a center wall **124** that extends between the top plate **108** and the bottom plate **112**. Brace walls **126** angle between the top plate **108** and the bottom plate **112**. The brace walls **126** exemplarily connect to the top plate **108** and/or the bottom plate **112** at three locations, at the respective side walls (e.g. tongue wall **116** and groove wall **120**), at the center wall **124**, and at a mid-point **125** between the center wall **124** and the respective tongue wall **116** or the groove wall **120**. In one example, the beam **102** is 1.2" high and 9" wide. As shown in FIG. 2, the brace walls **126** connect to the bottom plate **112** at the tongue wall, **116**, the center wall **124**, and at the groove wall **120**, while the brace walls **126** connect to the top plate **108** at the midpoints between the center wall **124** and the respective tongue wall **116** and groove wall **120**. In such an example, brace walls **126** divide the beam **102** into quarters and engage either the top plate **108** or the bottom plate every 2.25". In this manner the brace walls **126** form triangles within the interior of the beam **102** between either the top plate **108** or bottom plate **112** and two of: tongue wall **116**, groove wall **120**, center wall **124**, and brace walls **126**.

(20) FIG. 4A is a detailed view of a portion of the top plate **108** of the beam **102** as denoted by line 4-4 in FIG. 3. FIG. 4A thus depicts a portion of the top plate **108** in magnified detail. The top plate **108** includes grooves **150** into the surface of the top plate **108**. The grooves **150** may, for example, be constructed by extrusion or by machining. The grooves **150** define the ribs **110**. The ribs **110** include overhangs **152** which each define dovetails **154** of the grooves **150**.

(21) FIG. 4B, depicts the top plate **108** of FIG. 4A with a coating of a texture material **156**. In an exemplary embodiment the texture material is an epoxy combined with a grit media. In another example the texture material is an adhesive combined with a grit media. In an example, the grit media is coal slag. In another example, the grit media is silicon dioxide, or silica. The grit media is entrained in the epoxy to exemplarily form a slurry. The combined grit media and epoxy slurry is spread over the top plate and flows into the grooves **150** of the top plate **108** and into the dovetails **154**. As the texture material **156** solidifies from slurry form to solid form, the texture material **156** within the dovetails **154** forms a mechanical connection between the texture material **156** and the

top plate **108** to provide the textured surface **106**. In another example, the adhesive/epoxy portion of the texture material is first spread onto the top plate **108**, and then the grit media is placed into the adhesive/epoxy portion of the texture material before the adhesive/epoxy portion dries/cures, the adhesive/epoxy portion is thus secured to the grit media and to the top plate **108** to form the textured surface **106** in the manner previously described.

(22) FIG. **5** is a close-up detail of two beams **102A**, **102B** connected to one another. It can be seen from FIG. **5** that the beams **102A**, **102B** are connected along their lengthwise dimension by engagement between lateral walls. That is groove wall **120A** engages with tongue wall **116B**. The tongue **114B** is received within the groove **118A**. This serves two functions within the apparatus **100**. First, it aligns the beams **102A**, **102B** during assembly and helps to keep the beams **102A**, **102B** in alignment as a bead **128** is welded between the beams **102A** and **102B** within the chamfered corners **122A**, **122B**. Secondly, once assembled, the overlap of portions of the adjacent beams with the tongue **114B** within the groove **118A** transfers load between the beams to improve the overall strength of the apparatus **100** and to resist twist between the different beams **102** of the apparatus **100**.

(23) FIGS. **6** and **7** provide respective examples of end pieces **103**, **104** as may be used with the beams **102** to form the apparatus **100**. FIG. **6** is a side view of an example of a foot end **103** and FIG. **7** is a side view of an example of a hook end **104**. The foot end **103** includes an extension **136**, in this case in the shape of a wedge, which for example is configured to provide a transition from the ground to the beam **102** in the apparatus **100**. The hook end **104** includes an extension **138**, in this case having a single arcing shape, which is configured to engage an elevated position, for example a vehicle bed, loading dock, or stair landing. The extension **136** may further include ribbing **135** or other texture across the surface of the extension **136**. In another example, the surface of the extension **136** may include the grit surface as previously described. The foot end **103** further includes a shelf **130** configured to engage the bottom plate **112** of each beam **102**. An end cap **134** extends from the shelf **130** and is configured to extend across the open end of the beams **102**. A lip **132** extends from the extension in the region about the end cap **134**. The lip **132** extends in generally the same direction as the end cap **134**. A chamfer **131** provides the transition from the lip **132** to the end cap **134**. The foot end **103** may be configured to secure to the one or more beams **102** by friction or interference fit but may also be secured to the beams by adhesive, welding, rivets, crimping, or other manners of fastening as will be recognized from the present disclosure. In an example, a weld bead (not depicted) may extend across the foot end **103** and the beams **102** along the chamfer **131**.

(24) FIG. **8** depicts a still further example of the apparatus **100**. While like reference numerals are used to denote like features between, for example, FIGS. **1** and **8**, it will be recognized that other features between these examples or other examples disclosed herein may be combined within the scope of the present disclosure. In the apparatus **100** of FIG. **8**, the textured surface extends across each of the beams **102**. Additionally, the apparatus **100** includes two different end pieces **158**, **160**. The two different constructions of the end pieces enable the apparatus **100** to lie flat relative to the bottom plate **112** with both end pieces providing ramped surfaces from the plane of the bottom plate **112** to the plane of the top plate **108**. This, for example, enables the apparatus **100** to be effectively used as a bridge across a gap, for example between a trailer bed and a loading dock or platform. The different construction however further enables the apparatus **100** to better engage an elevated surface when the apparatus **100** is used as a ramp.

(25) FIGS. **9-11** depict another example of the apparatus **200** of the present disclosure, the apparatus **200** takes the form of a ramp, but which may also exemplarily be used as a load leveler for a step-deck trailer. It will be recognized that the apparatus **200** as disclosed herein is merely exemplary and features of apparatus **200** may be combined with features of apparatus **100** or may include fewer features than as shown and described with respect to apparatus **100** and apparatus **200** while remaining within the scope of the present disclosure. Referring to FIG. **9**, the apparatus

200 has a first end **201**, an opposite second end **202**, a top side **203**, a bottom side **204**, and a pair of opposing lateral sides **205**, **206**. A plurality of cross treads **208** are connected to the top side **203** with adhesives or welds, and a retention tube **211**, **212** is coupled at each end **201**, **202**. As explained, for example, in U.S. Pat. No. 8,347,439, previously incorporated by reference, the retention tubes **211**, **212** may be used to connect to a semi-trailer in either a ramp configuration or in a load leveler configuration. Openings **213** in the lateral sides **205**, **206** provide handles for use by an operator when moving the apparatus **200**, for example to position the apparatus into the previously described ramp configuration or load leveler configuration.

(26) The apparatus **200** is constructed of one or more extruded beams **220** that extend continuously between the first end **201** to the second end **202**. The beams **220** in an example wherein the apparatus includes more than one beam **220** (as depicted in FIG. 9, the beams **220** engage and are coupled to one another along their length dimension.

(27) FIG. 10 is a cross-sectional view of the apparatus **200** taken along line 10-10 of FIG. 9. Each beam **220** includes opposing side walls **221**. The side walls extend between a top plate **222** and a bottom plate **223**. The opposing side walls **221** intersect the bottom plate **223** at corners **225** and the side walls **221** intersect the top plate **222** at corners **225**. In the depicted exemplary embodiment, two adjacent beams **220** are coupled together lengthwise such that one of the side walls **221** of each beam **220** abuts the other. Preferably, the abutting side walls **221** are welded to each other.

However, the abutting side walls **221** can be coupled together by other means, such as adhesives and/or mechanical fasteners (e.g., nuts and bolts, rivets). The abutting side walls **221** form a “double-wall” at the center of the beam **220** that, when the side walls **221** are secured to each other, increases the structural rigidity of the apparatus **200**. When so connected, the combined beams **220** are structurally similar to the beam **220** described above. The connected side walls **221** effectively act as a combined, thickened, center wall (e.g. **124**).

(28) Brace walls **226** inwardly extend from the corners **225** and intersect with the top plate **222** such that the brace walls **226** divide the rectangular cavity of the beam **220** into separate triangular cavities **227**. The brace walls **226** connect to the top plate **222** at a midpoint between the side walls **221** of the beam **220**. In an example, the beam **220** has a width dimension W that is twice the height dimension H , the brace walls **226** form 45 degree angles from the respective corners **225** to the top plate **222**. The brace walls **226** thus meet and form a 90 degree interior angle. The brace walls **226** increase the structural rigidity of the beam **220** by supporting the top plate **222** and distributing weight applied to the top plate **222** or the side walls **221**. Thus, the brace walls **226** increase the weight capacity of the apparatus **200** when the apparatus **200** is in either the ramp configuration or the load leveler configuration as previously described.

(29) Referring back to FIG. 9, the apparatus **200** differs from the apparatus **100** in that the apparatus **200** includes truncated section **216** at end **201** and truncated section **217** at end **202**. The truncated sections include at least one angled dimension across the one or more beam **220** of the apparatus **200**. The beams **220** include a main section **M1** between the two truncated sections **216**, **217**. The cross-section of each beam **220** is generally constant lengthwise along the main section **M1**. The side walls **221** have a height $H1$ along the main section **M1**. The bottom plates **223** of each beam **220** are also co-extensive with the main section **M1**. The side walls **221** taper in the height dimension in a direction away from the bottom plate **223** towards the top plate **222**.

Respective end plates **229**, **230** are secured across the open portion of the beam **220** between the bottom plate **223** and the top plate **222**. In an example, the end hook **211** is secured to the end plate **229** to form an end cap **232** and the end hook **212** is secured to the end plate **230** to form an end cap **234**.

(30) FIG. 11 is a cross-sectional view taken along line 11-11 of FIG. 9 to provide an example of a cross-section through the transition section **216**. As shown in FIG. 11, the height $H2$ of the side walls **221** and that of the beams **220** as a whole is less than the height $H1$ of the side walls **221** of the beams **220** in the main section **M1**, as shown in FIG. 10. Furthermore, it can be seen that the

brace walls **224** retain the same angular arrangement as shown in FIG. **10**, but with the decreased height **H2** between the top plate **222** and the end plate **229**, the brace walls connect to the end plate **229** interior from side walls **221**. In examples, the brace walls **224** may be secured to the end plate **229** by adhesive or another known manner of securement, while in other examples, the brace walls **224** movably rest upon the end plate **229**. The end plate **229** may be secured to the beams by welding at the side walls, bottom plate **223** and the top plate **222**. In further examples, a single end plate **229**, **230** of a respective end cap **232**, **234** extends across more than one beam **220**.

(31) The top plates **222** of the beams **220** continuously extend along the apparatus **200** from the first end **201** to the second end **202**. That is, the top walls **222** extend along the main section **M1** and the truncated sections **216**, **217**. Thus, when in the ramp configuration, the top walls **222** of the beams **220** of the apparatus **200** form a continuous surface in both a width dimension and in a length dimension. In an example, the top plates **222** of the beams **220** are provided with a continuous surface that extends across all beams. The continuous surface may be a grit surface as described above with respect to apparatus **100**. In further examples, the continuous surface may be a punched or textured metal sheet that extends across one or more beams. In still other examples, cross treads **208** may be secured across one or more beams

(32) The beams **220** are formed with a non-corrosive metal, such as aluminum. Each beam **220** is formed as an extrusion. In the exemplary embodiment depicted, the apparatus **200** at least two beams **220** that are coupled together lengthwise in a side-by-side configuration. Thus, the apparatus **200** is formed with fewer individual members and welds than conventional combination ramp and load levelers. The apparatus **200** also has several other advantages over conventional combination ramp and load levelers including comparable strength with less metal, and thus lower overall weight per load capacity. In other examples, strengths, e.g. load capacity, may be achieved with an apparatus that is smaller in the vertical dimension compared to the aforementioned existing apparatus.

(33) It will be recognized that the apparatus **100** and the apparatus **200** are but examples of the present disclosure and that features of these examples may be combined, removed, or interchanged between the examples while remaining within the scope of the present disclosure. The apparatus **300** may be used in the ramp configuration or in the load leveler configuration as described above, although for the present description, the ramp configuration will be used.

(34) The apparatus and components thereof as described herein find numerous advantages over prior designs. The brace walls and beams that form a continuous apparatus surface provide improve strength in both a ramp configuration and a load leveler configuration. The brace walls help to transfer loads evenly across the beams and from the top plates of the beams to the bottom plates of the beams through the lateral walls, center walls, and brace walls.

(35) In examples, the strength of the truss may require all of the separate beams work together, being solidly joined to one another. While welding may be needed to connect the beams, welding can weaken the beam material by annealing the metal. Therefore, welding within a channel formed by chamfered corners between the beams provides a space for the welding bead and a comparatively smooth weld surface. Exemplary tongue and groove joints maintain beam alignment while improving strength, load transfer, and resistance to twisting.

(36) Further a system for ramp and/or load leveler construction that uses symmetry in design to provide two components e.g. beams and hooks which can be cut to dimensions and assembled to provide a variety of sizes and configurations of ramps and/or load levelers. In examples both the beam and the hook can be manufactured as extrusions and the components cut to the dimensions as needed. The configurations and designs disclosed herein provide the advantage of a ramp and/or load leveler construction with the same or greater strength with the same or less material, weight, and manufacturing time or cost.

(37) Citations to a number of references are made herein. The cited references are incorporated by reference herein in their entireties. In the event that there is an inconsistency between a definition

of a term in the specification as compared to a definition of the term in a cited reference, the term should be interpreted based on the definition in the specification.

(38) In the present description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. The different apparatuses, systems, and method steps described herein may be used alone or in combination with other apparatuses, systems, and methods. It is to be expected that various equivalents, alternatives and modifications are possible within the scope of the appended claims.

(39) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to make and use the invention. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A ramp, the ramp comprising: an extruded beam extending between a first end and a second end, the beam having: a first plate extending from the first end to the second end, wherein the first plate is a top plate; a second plate opposite the first plate and extending from the first end to the second end, wherein the second plate is a bottom plate; a first side wall and a second side wall that each extend from the first end to the second end between the first plate and the second plate; a first brace wall that extends from the bottom plate to the top plate and is angled from the first side wall interiorly in the direction of the second side wall; and a second brace wall that extends from the bottom plate to the top plate and is angled from the second side wall interiorly in the direction of the first side wall, wherein the first and second brace walls meet at a midpoint of the top plate; a first end piece secured about the first end and engaging at least one of the first plate or the second plate; and a second end piece secured about the second end and engaging at least one of the first plate or the second plate.
2. The ramp according to claim 1, wherein the first end is orthogonal to the first plate and the second end is orthogonal to the first plate.
3. The ramp according to claim 1, wherein the first end piece is a first end hook and the second end piece is a second end hook.
4. The ramp of claim 1, wherein the first side wall comprises a tongue extending lengthwise along the first side wall outwardly away from the second side wall and the second side wall comprises a groove extending lengthwise along the second side wall.
5. The ramp of claim 4, wherein the tongue is of a first dimension to fit within a second dimension of the groove.
6. The ramp according to claim 1, further comprising a textured surface secured to the first plate.
7. The ramp according to claim 1, wherein the first plate comprises grooves that define ribs and the grooves further define dovetails into the ribs.
8. The ramp according to claim 7, further comprising a textured surface secured to the first plate, wherein the textured surface comprises an adhesive or an epoxy, and the adhesive or epoxy solidifies within the dovetails to form a mechanical connection between the textured surface and the first plate.
9. A ramp, the ramp comprising: an extruded beam extending between a first end and a second end, the beam having: a first plate extending from the first end to the second end; a second plate opposite the first plate and extending from the first end to the second end; a first side wall and a second side wall that each extend from the first end to the second end between the first plate and

the second plate; a first brace wall that extends from a first corner to the first plate, the first corner being between the second plate and the first side wall; and a second brace wall that extends from a second corner to the first plate, the second corner being between the second plate and the second side wall; a first end piece secured about the first end and engaging at least one of the first plate or the second plate; a second end piece secured about the second end and engaging at least one of the first plate or the second plate; and a center wall that extends between the first plate and the second plate, wherein the center wall is equidistant between the first side wall and the second side wall.

10. The ramp of claim 9, wherein the center wall is parallel to the first side wall and to the second side wall.

11. The ramp of claim 9, further comprising: a third brace wall that connects to the first plate; and a fourth brace wall that connects to the first plate.

12. The ramp of claim 11, further comprising: the third brace wall extends from a third corner to the first plate, the third corner being between the center wall and the second plate; the fourth brace wall extends from a fourth corner to the first plate, the fourth corner being between the center wall and the second plate.

13. The ramp of claim 12, further wherein the first brace wall connects to the third brace wall at the first plate and the second brace wall connects to the fourth brace wall at the first plate.

14. A ramp comprising: first and second extruded beams that each extend between a first end and a second end, the beams each having: a first plate, the first plate having an exterior textured surface; a second plate opposite the first plate; a groove wall extending between the first plate and the second plate, the groove wall comprising a groove extending into the groove wall lengthwise along an outside of the groove wall; a tongue wall extending between the first plate and the second plate, the tongue wall comprising a tongue extending lengthwise along the first side wall outwardly from the tongue wall in a direction away from the groove wall; a first brace wall that extends from the second plate to the first plate; and a second brace wall that extends from the second plate to the first plate; a first end piece that engages across the first ends of the first and second extruded beams; a second end piece that engages across the second ends of the first and second extruded beams; wherein the tongue of the first extruded beam is received within the groove of the second extruded beam to connect the first extruded beam to the second extruded beam in a lengthwise manner; wherein the first extruded beam comprises a first chamfered exterior corner where the tongue wall of the first extruded beam connects to the second plate of the first extruded beam, and the second extruded beam comprises a second chamfered exterior corner where the groove wall of the second extruded beam connects to the second plate of the second extruded beam, and a weld joint secures the first extruded beam to the second extruded beam, the weld joint located between the first chamfered exterior corner and the second chamfered exterior corner.

15. The ramp of claim 14, wherein the first and second extruded beams each further comprise: a center wall that extends between the first plate and the second plate, wherein the center wall is equidistant between the groove wall and the tongue wall; a third brace wall that extends from a first corner to the first plate, the first corner being between the center wall and the second plate; a fourth brace wall that extends from a second corner to the first plate, the second corner being between the center wall and the second plate; wherein the first brace wall connects to the third brace wall at the first plate and the second brace wall connects to the fourth brace wall at the first plate.

16. The ramp of claim 14, wherein the first end piece comprises a first shelf that engages the second plates of the first and second extruded beams and the second end piece comprises a second shelf that engages the second plates of the first and second extruded beams.

17. The ramp of claim 14, wherein the exterior textured surfaces of the first and second extruded beams is a continuous textured surface across the respective first plates of the first and second extruded beams.
